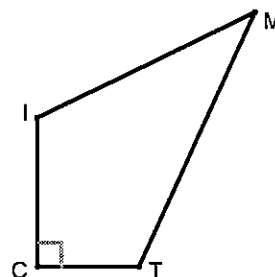


- Let k be the smallest positive solution for x for the equation $\sin x = 0.6$ where x is measured in degrees. Let w be the smallest positive solution for y for the equation $\sin y = 0.6$ where y is measured in radians. Determine the product (kw) . Express your answer as a decimal rounded to four significant digits.
- The minimum value of $f(x) = A \sin(2x + \pi) + 3$, where $A > 0$, is -7 . Determine the value of A .
- $\tan \theta = -\frac{7}{4}$, $\frac{\pi}{2} < \theta < \frac{3}{2}\pi$. Then $\sin \theta = \frac{k\sqrt{w}}{p}$ in reduced form with k , w , and p integers and $p > 0$. Determine the sum $(k + w + p)$.
- $\begin{cases} x = 4t + 5 \\ y = 5t - 1 \end{cases}$. Determine the slope of the line perpendicular to the graph of this parametric function when $t = -2$.
- For all permissible values of θ , $\frac{10 \sin^{(4)}(\theta) - 17 \sin^{(2)}(\theta) + 6}{18 - 15 \sin^{(2)}(\theta)} = k(\cos(2\theta))$. Determine the value of k . Express your answer as an integer or as a common or improper fraction.

- In quadrilateral $ICTM$, $\angle ICT = 90^\circ$ and $\angle CTM = 115^\circ$. Also, $IC = 8$, $CT = 6$ and $TM = 11$. Determine the numeric area of quadrilateral $ICTM$. Express your answer as a decimal rounded to the nearest thousandths.



- $f(x) = \frac{ax^2 - 11x + 5}{4 - 11x + 6x^2}$ has a horizontal asymptote at $f(x) = \frac{1}{3}$. Also, b represents the number of distinct vertical asymptotes for the graph of $f(x)$ and c represents the greatest number that is not an element of the range of $f(x)$. Determine the value of $f(b) + c$. Express your answer as an integer or as a common or improper fraction.

8. Three integers k , w , and p with $k < w < p$ form a geometric sequence. The sum of these three numbers is 21 and the sum of the reciprocals of these three numbers is $\frac{7}{12}$. Determine these three numbers. Express your answer as the ordered triple (k, w, p) .
9. Andrew and Brent will take turns rolling two fair standard dice until one of two things happens: Andrew gets a sum of 6 or Brent gets a sum of 5. Determine the probability that Brent wins when Brent goes first. Express your answer as an integer or as a common fraction.
10. The period of the graph of $y = 4\sin x \cos x (\cos^2 x - \sin^2 x)$ can be expressed as $k\pi$. Determine the value of k . Express your answer as an integer or as a common or improper fraction.
11. The equation $y = 10e^{(-t)} \cos\left(\frac{t}{3}\right)$ represents a damped vibration, such as the displacement from equilibrium of the end of a stretched spring or the end of struck tuning fork. Determine the smallest value for t after which $|y|$ never exceeds ten per cent of its maximum value. Express your answer as a decimal rounded to the nearest thousandth.
12. $f(n+1) = \frac{3f(n)+1}{3}$ and $f(1) = 5$. Determine the value of $f(2019)$. Express your answer as an integer or as a common or improper fraction.
13. Aune and Ella are trying to meet at the library. Aune can arrive between 3 pm and 5 pm and stay for 17 minutes. Ella can arrive between 3:30 pm and 4:30 pm and stay for 34 minutes. Determine the probability they will be at the library at the same time. Express your answer as an integer or as a common fraction.
14. Let a be a positive integer chosen at random from the interval $[1, 100]$. Then let $k = \frac{a}{2} + \frac{a}{3} + \frac{a}{4} + \frac{a}{5} + \frac{a}{6} + \frac{a}{7}$. Determine the probability that when k is expressed as a simplified and reduced fraction, the numerator of k will be prime. Express your answer as an integer or as a common fraction.

15. Three vertices of a regular octagon are used as vertices of a triangle ABC . The exact smallest possible ratio k for $\frac{\text{Area of triangle}}{\text{Area of octagon}}$ can be expressed as $k = \frac{a+b\sqrt{c}}{d}$ in reduced form with a, b, c , and d integers and $d > 0$. Determine the sum $(a+b+c+d)$.
16. Determine the remainder when $(x+1)^5 + (x+2)^4 + (x+3)^3 + (x+4)^2 + (x+5)$ is divided by $(x+2)$. Express your answer as an integer or as a common or improper fraction.
17. $i = \sqrt{-1}$ and $z = a+bi$, a and b real. When graphed in the complex plane, the roots of $z^{12} = 1$ determine vertices of a regular dodecagon. The perimeter of this regular dodecagon can be expressed as $k = m\sqrt{n} + p\sqrt{q}$, m, n, p , and q integers. Determine the sum $(m+n+p+q)$.
18. A set of points in the xy -plane consists of all those points satisfying the inequality $x^2 + y^2 < 1$. The graph of this region has numeric area Z . A curve with the equation $\frac{1}{x} + \frac{1}{y} = \frac{1}{xy}$, where $xy \neq 0$, separates the region into two parts. The numeric area A of the larger part can be represented as $A = \frac{k\pi + w}{p}$ in reduced form with k, w , and p integers and $p > 0$. Determine the sum $(k+w+p)$.
19. The sum of all values for x in the interval $[0, 2\pi]$ such that $\sqrt{3}\sin x + \cos x = \sqrt{2}$ is $k\pi$. Determine the value of k . Express your answer as an integer or as a common or improper fraction.
20. The foci of a graph of an ellipse $\frac{x^2}{k} + \frac{y^2}{w} = 1$ are located at $(0, \sqrt{5})$ and $(0, -\sqrt{5})$. The major axis has length 6. Determine the sum $(k+w)$.

2019 RAA

Name ANSWERS

Pre-Calculus

School _____

(Use full school name – no abbreviations)

_____ Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

- | | |
|---|--|
| 1. <u>23.73</u> (Must be this decimal.) | 11. <u>2.049</u> (Must be this decimal.) |
| 2. <u>10</u> | 12. <u>$\frac{2033}{3}$</u> (Must be this reduced improper fraction.) |
| 3. <u>137</u> | 13. <u>$\frac{763}{1800}$</u> (Must be this reduced common fraction.) |
| 4. <u>$-\frac{4}{5}$ OR $-\frac{-4}{5}$</u> (Must be this reduced improper fraction.) | 14. <u>$\frac{11}{100}$</u> (Must be this reduced common fraction.) |
| 5. <u>$\frac{1}{3}$</u> (Must be this reduced common fraction.) | 15. <u>11</u> |
| 6. <u>72.503</u> (Must be this decimal.) | 16. <u>7</u> |
| 7. <u>$\frac{29}{5}$</u> (Must be this reduced improper fraction.) | 17. <u>8</u> |
| 8. <u>(3,6,12)</u> (Must be this ordered triple.) | 18. <u>9</u> |
| 9. <u>$\frac{9}{19}$</u> (Must be this reduced common fraction.) | 19. <u>$\frac{2}{3}$</u> (Must be this reduced common fraction.) |
| 10. <u>$\frac{1}{2}$</u> (Must be this reduced common fraction.) | 20. <u>13</u> |

1. A boy travels along a path given by $r = 5 \sin \theta$ where r is measured in feet. Then the distance in feet this boy traveled for $0 \leq \theta \leq 2\pi$ is $k\pi$. Determine the value of k .
2. An owl was perched on a vertical pine tree with his eyes 24 feet above ground. This owl spots a mouse on the level ground at an angle of depression of 50° . Determine the straight-line distance from the owl to the mouse. Express your answer rounded to the nearest tenth of a foot.
3. (k, w) is the solution to the matrix equation $\begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Determine the value of the sum $(k + w)$.
4. Let $k = 8 \left[\sum_{n=1}^{\infty} \left(\frac{1}{3} \right)^n \right]$. Determine the value of k .
5. Let $W(\theta) = (\cos \theta, \sin \theta)$ be the wrapping function around the unit circle. Determine the straight-line distance between $W(\pi)$ and $W\left(\frac{\pi}{4}\right)$. Express your answer as a decimal rounded to the nearest thousandth.
6. Two cruise ships leave the same port at the same time so that the angle between their paths is 60° . The first ship travels at 16 mph and the second ship travels at 14 mph. Determine how long they had been sailing at the time they are 26.424 miles apart. Express your answer as a decimal rounded to the nearest tenth of an hour.
7. Two lines $y_1 = \frac{1}{4}x - 3$ and $y_2 = 5x + \frac{1}{2}$ intersect. Determine the degree measure of the obtuse angle formed by this intersection. Express your answer as a decimal rounded to the nearest tenth of a degree.

8. Let vectors $\vec{u} = -6\vec{i} - 78\vec{j}$ and $\vec{v} = 117\vec{i} + k\vec{j}$ such that $\vec{u} \perp \vec{v}$. Determine the value of k . Express your answer as an integer or as a common or improper fraction.

9. **(Always, Sometimes, or Never)** For your answer, write the whole word **Always**, **Sometimes**, or **Never**—whichever is correct.

The graph of a certain linear function has a negative slope. Consequently, if its x -intercept is a negative number, then its y -intercept is also a negative number.

10. Let $\sin x = \frac{1}{3} + \cos x$. Determine the value of $\sin(2x)$. Express your answer as an integer or as a common or improper fraction.

11. Box A contains 3 red marbles and 6 blue marbles. Box B contains 2 red marbles and 1 blue marble. Box C contains 2 red marbles and 4 blue marbles. Two of the boxes are randomly chosen. Then one marble is chosen from each of those two boxes. Determine the probability that these two marbles drawn are the same color. Express your answer as a common fraction.

12. Consider the graph of the function $f(x) = 3\sin(2x + \pi) - 5$. Determine the amplitude of $f(x)$.

13. Let x represent the degree measure of an angle such that $\tan(x) = \sqrt{3}$. If $0^\circ < x^\circ < 720^\circ$, determine the sum of all possible distinct values of x .

14. a , b , and c are positive integers with $b < c$ such that $\frac{1}{\sqrt[3]{2}-1} = a + \sqrt[3]{b} + \sqrt[3]{c}$. Determine the ordered triple (a, b, c) . Express your answer as that ordered triple.

15. Line a has slope $\frac{3}{2}$ and line b has slope 4. Line ℓ bisects an angle formed by a and b and has positive slope. That slope can be written as $\frac{k+w\sqrt{p}}{q}$ where k , w , and q are relatively prime integers and $q > 0$. Determine the sum $(k+w+p+q)$.
16. Let T be the tetrahedron bounded by the points $(0,0,0)$, $(0,0,8)$, $(0,6,0)$, and $(10,0,0)$. The unit vector $\vec{V}$ normal to the face of the tetrahedron that contains the points $(0,0,8)$, $(0,6,0)$, and $(10,0,0)$ can be written as $\vec{V} = \left\langle \frac{x}{\sqrt{w}}, \frac{y}{\sqrt{w}}, \frac{z}{\sqrt{w}} \right\rangle$ with $x > 0$. Determine the sum $(x+y+z+w)$.
17. Let $p(x)$ and $q(x)$ be quartic polynomial functions with integral coefficients such that $p(1)=q(3)$, $p(2)=q(4)$, $p(3)=q(5)$, $p(4)=q(6)$, and $p(5)=q(7)+k$, where k is a positive multiple of 23. Determine the smallest value of k that satisfies these conditions.
18. In $\triangle ABC$ with right angle at C , $\frac{AC}{BA} = \frac{56}{65}$. $\tan \angle BAC = k$. Determine the value of k . Express your answer as an integer or as a common or improper fraction reduced to lowest terms.
19. $\lim_{x \rightarrow \infty} \frac{4x + e^{(-x)}}{5x} = k$. Determine the value of k . Express your answer as an integer or a common or improper fraction.
20. A plane flying at an average speed of 420 mph at an angle of 40° west of due north encounters a wind pushing 20 mph at an angle of 20° west of due south. The plane will then be traveling at a new angle west of due north. Determine the degree measure of that angle. Express your answer as a decimal rounded to the nearest hundredth.

2018 RAA

Name ANSWERS

Pre-Calculus

School _____

(Use full school name – no abbreviations)

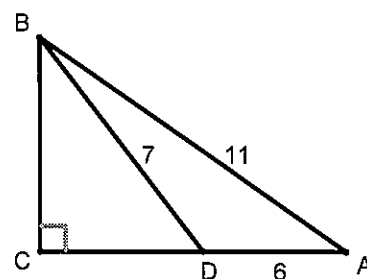
_____ Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

- | | |
|---|---|
| 1. <u>10</u> ("feet" or "ft." optional.) | 11. <u>$\frac{13}{27}$</u> (Must be this reduced common fraction.) |
| 2. <u>31.3</u> (Must be this decimal, "feet" or "ft." optional.) | 12. <u>3</u> |
| 3. <u>-1</u> | 13. <u>1320</u> ("degrees" or "°" optional.) |
| 4. <u>4</u> | 14. <u>(1,2,4)</u> (Must be this ordered triple.) |
| 5. <u>1.848</u> (Must be this decimal.) | 15. <u>243</u> |
| 6. <u>1.7</u> (Must be this decimal, "hours" or "hr." optional.) | 16. <u>816</u> |
| 7. <u>115.3</u> ("degrees" or "°" optional.) | 17. <u>552</u> |
| 8. <u>-9</u> | 18. <u>$\frac{33}{56}$</u> (Must be this reduced common fraction.) |
| 9. <u>Always</u> (Must be this whole word.) | 19. <u>$\frac{4}{5}$</u> (Must be this reduced common fraction.) |
| 10. <u>$\frac{8}{9}$</u> (Must be this reduced common fraction.) | 20. <u>42.42</u> ("degrees" or "°" optional.) |

1. Let $k = \sin\left(\frac{2}{3}\pi\right)$ and $w = \cot\left(\frac{5}{6}\pi\right)$. Determine the exact product (kw) .
2. $\text{Arc sin } x$ represents the inverse sine function. Determine the exact value of $\text{Arc sin}\left(\cos\left(\frac{7}{6}\pi\right)\right)$.
3. Let $f(x) = 2x^2 - 5x - 9$. Then $\lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right) = kx^2 + wx + p$. Determine the value of $(k + w + p)$.
4. A particular triangle has sides of length 20 and 17 and an included angle between those two sides of measure 52° . Determine the area of this triangle. Express your answer as a decimal rounded to the nearest thousandth.
5. Let $k = \sum_{n=0}^{\infty} (18) \left(\frac{17}{20} \right)^n$. Determine the exact value of k .

6. $\triangle ABC$ is a right triangle with right angle at C . $AB = 11$, $BD = 7$, and $DA = 6$. Determine the length of $\overline{BC}$. Express your answer as a decimal rounded to the nearest thousandth.



7. Let matrices $A = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 7 & 3 \\ 3 & 2 \end{bmatrix}$. Determine the value of $\det(A \times B)$ (the determinant of the product of matrices A and B .)

8. A Super Ball is launched 100 feet into the air from ground level, measured vertically to the bottom of the ball. It falls back to ground level and then rebounds $\frac{3}{4}$ of its height on the previous bounce. The ball continues this pattern until it stops. Determine the total distance, in feet, the ball traveled as measured from ground level vertically to the bottom of the ball.
9. The equation $(y - 2)^2 = 5(x + 3)$ for the standard coordinate system is rewritten in terms of a coordinate system (x', y') for which the new origin is $(5, 2)$ with axes parallel to the original xy - axes. The resulting equation can be written in the form: $(y')^2 = k(x' + w)$ where k and w are positive integers. Determine the value of $(k + w)$.
10. Assume the following is NOT a degenerate form of a conic. Identify $6x^2 + 5xy + 2y^2 + x + 2y - 3 = 0$ as a circle, ellipse, hyperbola, or parabola. Express your answer by writing the whole word for this conic.
11. $i = \sqrt{-1}$. Complex $z = 1 + i\sqrt{3}$. One square root of z is $\sqrt{z} = a + bi$ with a and b exact real numbers and in simplified radical form and with $a > 0$. Determine the ordered pair (a, b) . Express your answer as the ordered pair (a, b) with exact entries.
12. Twins Jordan and Michael have the following growth curves, with x measured in years and y in inches: $y_1 = 20 + 54\left(1 - e^{\left(\frac{-x}{3}\right)}\right)$ and $y_2 = 21 + 53\left(1 - e^{\frac{-x}{3.1}}\right)$. Determine the sum of all the year(s) after birth at which their heights were equal. Express your answer as a decimal rounded to the nearest thousandth.
13. The rectangular coordinates $(-2, -2\sqrt{3})$ can be written in polar form as (r, θ) where $0 \leq \theta < 2\pi$ in radian measure. Determine these polar coordinates. Express your answer as the ordered pair (r, θ) .
14. In $\triangle ABC$, $\angle BAC = 74^\circ$ and $AC = 100$. $\overline{BC}$ has integral length and is shorter than $\overline{AC}$. Determine the sum of all possible integral lengths for $\overline{BC}$.

15. Today is Aune's fifteenth birthday. Her parents measure her height ("for the last time!") to be 62.5 inches. On her fourteenth birthday her height was 61.0 inches. They noted her height in inches has been increasing geometrically for the past five years. Determine Aune's height, in inches, on her tenth birthday. Report your answer as a decimal rounded to the nearest quarter of an inch.
16. Let $f(x) = 3 + 4\cos(x + \pi)$ with x measured in radians. Let k represent the amplitude of the graph of $f(x)$ and w represent the y-intercept of the graph of $f(x)$. Determine the sum $(k + w)$.
17. In $\triangle ABC$, D lies on $\overline{AC}$ such that $\overline{BD}$ bisects $\angle ABC$. $AB = 18$, $BC = 12$, and $DC = 7$. Determine the degree measure, in degrees, for the smallest angle of $\triangle ABC$. Express your answer as a decimal rounded to the nearest thousandth.
18. Let $\sin \theta = -\frac{1}{2}$ where $0 \leq \theta < 2\pi$ is measured in radians. The sum of all possible value(s) for θ is $k\pi$. Determine the value of k . Express your answer as an integer or as a common or improper fraction reduced to lowest terms.
19. When $2\sin(2x)\cos x - 4\cos^2 x = \sin x - 1$ is solved over $0 \leq x \leq 2\pi$, the sum of the solutions is $k\pi$. Determine the exact value of k . Express your answer as an integer or as a common or improper fraction reduced to lowest terms.
20. The equation of the directrix of a parabola with vertex $(-3, 2)$ that passes through $(9, -16)$ and opens down is either $x = k$ or $y = w$. Determine the equation of this directrix. Express your answer as the equation $x = k$ or $y = w$.

2017 RAA

Name ANSWERS

Pre-Calculus

School _____

(Use full school name – no abbreviations)

_____ Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

1. $-\frac{3}{2}$ OR $-\frac{3}{2}$ OR -1.5 OR $-1\frac{1}{2}$

2. $-\frac{1}{3}\pi$ OR $-\frac{\pi}{3}$ OR $-\frac{\pi}{3}$ (Must be this exact answer.)

3. -1

4. 133.962 (Must be this exact decimal.)

5. 120

6. 6.325 (Must be this exact decimal.)

7. 15

8. 800 ("feet" or "ft." optional.)

9. 13

10. ellipse (Must be the whole word, caps optional.)

11. $\left(\frac{\sqrt{6}}{2}, \frac{\sqrt{2}}{2}\right)$ (Must be this ordered pair with exact values.)

12. 1.738 (Must be this decimal.)

13. $\left(4, \frac{4}{3}\pi\right)$ OR $\left(-4, \frac{\pi}{3}\right)$ (Either exact coordinate pair accepted; only one.) answer needed)

14. 294

15. 55.25 (Must be this decimal, "inches" or "in." optional.)

16. 3

17. 39.482 (Must be this decimal, "degrees" or "°" opt.)

18. 3

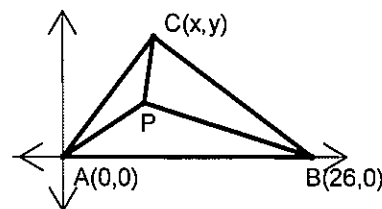
19. $\frac{9}{2}$ (Must be this reduced improper fraction.)

20. $y = 4$ (Must be this equation.)

1. Determine the value of the indicated vector inner (dot) product $(-5, 3) \cdot (6, 8)$.
2. Determine the value of $\lim_{x \rightarrow 0} \frac{5x^3 + 3x^2 - 7x + 3}{2x - 3}$.
3. Let $\tan \theta = \frac{1}{2}$ with θ in quadrant I. Then the exact value for $\cos \theta + \cot \theta = \frac{k + w\sqrt{p}}{q}$ in simplified and reduced radical form with k , w , p , and q integers. Determine the sum $(k + w + p + q)$.
4. Determine the value of the sum $\sum_2^5 (2^k - 1)$.
5. A sine curve has a maximum value of 3 at $x = \frac{4}{3}\pi$, and a minimum value of -3 at $x = \frac{10}{3}\pi$ with exactly one zero between these two values for x . Determine the standard equation $y = A \sin[B(x - C\pi)] + D$ for this curve with x measured in radians and where $A > 0$ is the least possible value. Express your answer as the sum $(A + B + C + D)$ written as a common or improper fraction reduced to lowest terms.
6. Let $f(x) = x^3 + \frac{3}{2}$, $g(x) = 2x\sqrt{x}$, and $h(x) = \sin^2 x$. Determine the value of $f\left(g\left(h\left(\frac{\pi}{6}\right)\right)\right)$. Express your answer as a common or improper fraction reduced to lowest terms.

7. Let $f(x) = (x-a)(x-b)^2(x+c)$ with $0 < a < b < c$. Determine the solution for $f(x) < 0$. Express your answer(s) in interval notation. (For example, the interval (k, w) represents $k < x < w$.)
8. In a geometric sequence, the last term is 1458, the common ratio is -3 , and the sum of the terms is 1094. Determine the second term of this geometric sequence.
9. Let a , b , c , and d be positive real numbers. Let $a+b+c+d=10$, let $c=d$, and let $a^2+b^2+c^2+d^2=30$. The largest possible value of c can be expressed as $\frac{k+\sqrt{w}}{f}$ in simplified and reduced radical form where k , w , and f are positive integers. Determine the value of $(k+w+f)$.
10. Determine the numeric area of a triangle with sides 20 and 16 and the included angle between those two sides having measure 150° .
11. In $\triangle ABC$, $2\sin A + 3\cos B = 4$ and $3\sin B + 2\cos A = 3$. Determine the degree measure of $\angle C$.
12. Points $(1, 3, -1)$, $(-2, 1, 5)$, and $(3, 2, 6)$ lie on a plane. A standard equation for that plane can be written as $Ax + By + Cz + D = 0$ where A , B , C , and D are relatively prime integers and $A > 0$. Determine the sum $(A + B + C + D)$.
13. A parabola has the equation of $2x^2 + 12x = 3y$. The equation of the directrix of this parabola can be expressed in the form $x = k$ or $y = w$. Determine the value of k or w , whichever is appropriate, as an exact decimal. Express your answer as the equation $x = k$ or $y = w$.

14. $(\sqrt{3} + \sqrt{5})$ is one zero of a fourth-degree polynomial with integer coefficients and leading coefficient 1 when written in decreasing degree of x . Determine the constant term for this polynomial.
15. Determine the number of distinct arrangements of the letters in the word MATHEMATICS.
16. The roots for x in $x^3 + ax^2 + bx - 105 = 0$ are each two more than the roots for x in $x^3 - dx^2 + ex - f = 0$. a , b , d , e , and f are constant real numbers. Determine the value of the sum $(4d + 2e + f)$.
17. For all real values of θ such that $\cos(\theta) + \sin(\theta) \neq 0$, $\frac{\cos^3(\theta) + \sin^3(\theta)}{\cos(\theta) + \sin(\theta)} = k - w(\sin(2\theta))$. Determine the value of $(7k + 28w)$.
18. Let $t_1 = 77$ and $t_{n+1} = \frac{1 - 24t_n}{24 - 3t_n}$ for $n \geq 1$ be a sequence. Determine the value of t_{2016} . Express your answer as a common or improper fraction reduced to lowest terms.
19. Determine the number of points of intersection for the curves $y = 1 - \frac{x}{4}$ and $y = \frac{\sin x}{x}$ where x is measured in radians.
20. In the diagram with coordinates as shown, $AC = 51$ and $BC = 55$. The ratio of the area of $\triangle PAB$ to the area of $\triangle PAC$ to the area of $\triangle PBC$ is 1:2:3. Determine the **ordered pair** that represents point P. Express your answer as an **ordered pair** (k, w) with **each member** of the ordered pair expressed as a common or improper fraction reduced to lowest terms.



2016 RAA

Pre-Calculus

Name ANSWERS

School _____

(Use full school name – no abbreviations)

_____ Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

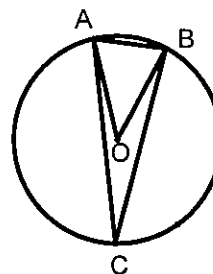
- | | |
|--|--|
| <p>1. <u> -6 </u></p> <p>2. <u> -1 </u></p> <p>3. <u> 22 </u></p> <p>4. <u> 56 </u></p> <p>5. $\frac{23}{6}$ (Or any value of the form $\frac{23}{6} \pm 4n$ where n is an integer)</p> <p>6. $\frac{97}{64}$ (Must be this improper fraction.)</p> <p>7. $(-c, a)$ (Must be only this ordered pair used as interval notation.)</p> <p>8. <u> -6 </u></p> <p>9. <u> 12 </u></p> <p>10. <u> 80 </u></p> | <p>11. <u> 90 </u> ("degrees" or "°" optional.)</p> <p>12. <u> 52 </u></p> <p>13. <u> $y = -6.375$ </u> (Must be this equation in y using this exact decimal value.)</p> <p>14. <u> 4 </u></p> <p>15. <u> 4989600 </u> ("arrangements" and comma usage optional.)</p> <p>16. <u> 97 </u></p> <p>17. <u> 21 </u></p> <p>18. $\frac{1847}{207}$ (Must be this reduced improper fraction.)</p> <p>19. <u> 2 </u> ("points of intersection" or "points" optional.)</p> <p>20. <u> $\left(\frac{739}{78}, \frac{110}{13}\right)$ </u> (Must be this ordered pair with reduced improper fraction entries.)</p> |
|--|--|

1. Points $P = (-3, 4)$ and $Q = (-5, 2)$. Determine the exact magnitude of vector $\overrightarrow{PQ}$.
2. $f \circ g(x) = f(g(x))$. If $f(x) = 5x + 6$ and $g(x) = 3x - 1$, then $f \circ g(x) = kx + w$. Determine the ordered pair (k, w) .
3. Determine all complex roots for the equation $x^2 + 40 = -9$.
4. x and y are real numbers such that $x^2 + 3xy + y^2 = 60$. Determine the maximum possible value for the product (xy) .
5. The largest real value of x such that $\sin 3x + \sin x = 0$ when solved over the interval $[0, 2\pi)$ is $x = k\pi$. Determine the value of k . Express your answer as a common or improper fraction reduced to lowest terms.
6. Consider the graph of the function $y = \frac{1}{2}\cos(8\pi x - 4) + 12$. Let k represent the amplitude of this function and w represent its period. Determine the exact value of the product (kw) . Express your answer as a common or improper fraction reduced to lowest terms.
7. A sequence is defined by the rule $F_{n+1} = F_n + F_{n-1}$. $F_4 = 7$ and $F_7 = 29$. Determine the value of F_{10} .
8. Solve for all values of x such that $\log_{100}(x+2) = \log x$.

9. In vector notation, $\vec{v} = \langle a, b \rangle = a\vec{i} + b\vec{j}$. Determine the degree measure of the angle between the vectors $\vec{u} = \langle 2, 3 \rangle$ and $\vec{v} = \langle -2, 5 \rangle$. Express your answer as a decimal rounded to the nearest hundredth.

10. The probability a certain man will be alive 25 years hence is $\frac{3}{7}$ while independently, the probability his wife will be alive 25 years hence is $\frac{4}{5}$. Determine the probability that at least one of this man and woman is alive 25 years hence. Express your answer a common fraction reduced to lowest terms.

11. In the figure shown, the radius of Circle O is 8 and $AB = 6$. Determine the degree measure of $\angle C$. Express your answer as a decimal rounded to the nearest hundredth.



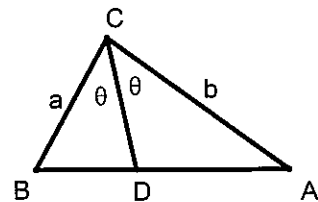
12. If $x > 5$, determine the largest value of k so that $\log(x) - k = \log(x-5) + 3$ has no real solutions.

13. $\frac{4x^4 + 2x^3 + 5x^2 + 15}{x+5} = Q(x) + \frac{k}{x+5}$. Determine the value of k .

14. $\angle A$ and $\angle B$ are first quadrant angles such that $\sin A = \frac{4}{7}$ and $\cos B = \frac{5}{7}$. In reduced and simplified radical form, $\tan(A+B) = \frac{k\sqrt{p} + w\sqrt{q}}{f}$ where k , w , p , q , and f are integers. Determine the value of the sum $(k + w + p + q + f)$.

15. Evie and Xavier are playing a game rolling a fair standard six-sided die. Evie wins if she rolls a 6 before Xavier rolls an odd number. If Evie rolls first, determine the probability Evie wins the game. Express your answer as a common fraction reduced to lowest terms.
16. Henrik measured the angle of inclination from the ground to the top of a vertical pole along a line on level ground and found the angle to be 50° . He walked 10 feet further from the pole along the same line and measured the angle of inclination from ground level to the top of the pole to be 40° . If he walked along the same line to a point 10 feet from the base of the pole, the angle of inclination from ground level to the top of the pole is k° . Determine k . Express your answer as a decimal rounded to the nearest hundredth of a degree.
17. Determine the point on the graph of the function $y = \sqrt{2x-5}$ that is closest to the point $(5,0)$. Express your answer as an ordered pair with exact coordinate entries.
18. When written out as an integer, $2015!$ has k trailing zeros (2015 factorial ends in a string of k zeros.) Determine the value of k .
19. Let $\sec \theta = \frac{20}{15}$. Determine the least possible exact value for $\sin \theta$.

20. In the diagram shown, $\overline{CD}$ bisects $\angle ACB$ in $\triangle ABC$.
 $\cos \theta = \frac{1}{4}$ and $\frac{1}{a} + \frac{1}{b} = \frac{1}{20}$. Determine the exact length CD .



2015 RAA

Pre-Calculus

Name ANSWERS

School _____

(Use full school name – no abbreviations)

_____ Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

- | | |
|---|--|
| <p>1. <u> $2\sqrt{2}$ </u> (Must be this exact answer.)</p> | <p>11. <u> 22.02 </u> (Must be this decimal, "degrees" or ° optional.)</p> |
| <p>2. <u> (15,1) </u> (Must be this ordered pair.)</p> | <p>12. <u> -3 </u></p> |
| <p>3. <u> $7i, -7i$ OR $0 + 7i, 0 - 7i$ </u> (Must have both answers in either form or order, but may use +/-.)</p> | <p>13. <u> 2390 </u></p> |
| <p>4. <u> 12 </u></p> | <p>14. <u> 62 </u></p> |
| <p>5. <u> $\frac{3}{2}$ </u> (Must be this reduced improper fraction.)</p> | <p>15. <u> $\frac{2}{7}$ </u> (Must be this reduced common fraction.)</p> |
| <p>6. <u> $\frac{1}{8}$ </u> (Must be this reduced common fraction.)</p> | <p>16. <u> 70.57° </u> (Must be this decimal, "degrees" or ° optional.)</p> |
| <p>7. <u> 123 </u></p> | <p>17. <u> $(4, \sqrt{3})$ </u> (Must be this exact ordered pair.)</p> |
| <p>8. <u> 2 </u> (This value only.)</p> | <p>18. <u> 502 </u> ("Zeros optional.)</p> |
| <p>9. <u> 55.49 </u> (Must be this decimal, "degrees" or ° optional.)</p> | <p>19. <u> $-\frac{\sqrt{7}}{4}$ OR $-\frac{1}{4}\sqrt{7}$ OR $-0.25\sqrt{7}$ </u> (Or exact reduced equivalent.)</p> |
| <p>10. <u> $\frac{31}{35}$ </u> (Must be this reduced common fraction.)</p> | <p>20. <u> 10 </u></p> |

1. One of the solutions for x in the equation $x^2 + bx + c = 0$ where b and c are integers is $x = 1 + \sqrt{2}$. Find the value of the sum $(b + c)$.
2. $i = \sqrt{-1}$. Solve for b if $1 - \frac{b}{4 + i\sqrt{3}} + \frac{b}{4 - i\sqrt{3}} = 0$.
3. The graph of $4x^2 - y^2 - 8x + 2y = 1$ is rotated 90° clockwise about the origin. Find the coordinates of the center of the new conic. Express your answer as the ordered pair (x, y) .
4. Find **the number** of polar points below that are coincident (occupy the same position) as the polar point $\left(5, \frac{\pi}{6}\right)$?
I. $\left(-5, \frac{7\pi}{6}\right)$ II. $\left(5, \frac{5\pi}{6}\right)$ III. $\left(5, \frac{13\pi}{6}\right)$ IV. $\left(5, \frac{11\pi}{6}\right)$ V. $\left(-5, \frac{-\pi}{6}\right)$
5. Find the value(s) of x for which $4 + 3\sqrt{7 - x} = x + 7$.
6. Simplify the expression $(\cos^6 x + 3(\cos^4 x)(\sin^2 x) + 3(\cos^2 x)(\sin^4 x) + \sin^6 x)$.
7. Let $P(x) = 0$ be a third degree polynomial equation with **integer** coefficients and leading coefficient of 1. Let the polynomial equation $P(x) = 0$ have at least one root that is an **integer**. If $P(6) = -11$, $P(4) = 3$, and $P(-4) = -341$, find $P(10)$.

8. A bag contains red and blue marbles. When two marbles are drawn, the probability that they are both red is equal to the probability they are both blue. The probability that one of each color is drawn is $\frac{4}{7}$. Find the least number of marbles that are in the bag.
9. The distance between the lines $y = \frac{3}{4}x + b$ and $3x - 4y = 7$ is 2. If $b > 0$, find the value of b .
10. If $i = \sqrt{-1}$, find the sum of an infinite geometric series whose first term is $2 - i$ and common ratio is $\frac{1}{2}i$.
11. Determine the numerical coefficient of $x^2y^2z^2$ in the simplified expansion of $(1+x)^5(1+y)^4(1+z)^3$.
12. The solutions for x in the equation $2(\ln x)^2 = \ln x^5 + 3$ are of the form e^a where e is the base of a natural logarithm. Find the sum of all possible values of a . Write your answer as a common or improper fraction reduced to lowest terms.
13. Given $f(4x+3) = 8x-1$, $f^{-1}(x) = ax+b$. Find the sum $(a+b)$.

14. $u = 2 \sin x + \frac{\cos^2 x}{2 \sin x + \frac{\cos^2 x}{2 \sin x + \frac{\cos^2 x}{2 \sin x + \dots}}}$, where $0 < \cos x < \sin x$ and $\cos x = \frac{1}{3}$. Find the exact value of u . Write your answer as a single reduced fractional expression.

15. Find the number of distinct real values for x so that $\begin{bmatrix} x-2 & x+5 \\ 2x & x-3 \end{bmatrix}$ will not have an inverse.

16. When David and Jim play pool, the probability that David wins exactly two games out of four is equal to the probability that he wins exactly three games out of four. Find the probability that Jim wins any given game. Write your answer as a common fraction reduced to lowest terms. Assume that the outcome of a game cannot affect the probabilities in other games, and that both players have a non-zero chance of winning a game.

17. A parabola is described by $\begin{cases} x = 2 + t \\ y = 1 - t^2 \end{cases}$. Find the exact distance between the vertex and the y -intercept of this parabola.

18. Determine the exact sum of all values of θ , $0 \leq \theta < 2\pi$ for which $\cos^2 \theta = \frac{3}{5}$.

19. The function $y = f(x)$ is periodic with period 3. If $f(6) = 4$, $f(8) = 2$ and $f(10) = 9$, find the value of $2f(1) + 3f(2) + 4f(3)$.

20. In a triangle with sides a , b , and c , $(a+b+c)(a+b-c) = 3ab$. Find $\sec C$, where angle C is the angle opposite side c .

2014 RAA

PreCalculus

Name ANSWERS

School _____

(Use full school name – no abbreviations)

_____ Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

1. -3

2. $\frac{19i\sqrt{3}}{6}$ (Must be this exact answer only
or exact algebraic equivalent.)

3. $(1, -1)$ (Must be this
ordered pair.)

4. 2 OR TWO

5. 3

6. 1

7. 177

8. 8

9. $\frac{3}{4}$ OR 0.75 OR .75

10. 2

11. 180

12. $\frac{5}{2}$ (Must be this reduced
improper fraction.)

13. 4

14. $\frac{2\sqrt{2}+3}{3}$ OR $\frac{3+2\sqrt{2}}{3}$ (Must be this exact
single fract'l expr.)

15. 2

16. $\frac{2}{5}$ (Must be this reduced
common fraction.)

17. $2\sqrt{5}$ (Must be this
exact answer.)

18. 4π (Must be this
exact answer.)

19. 40

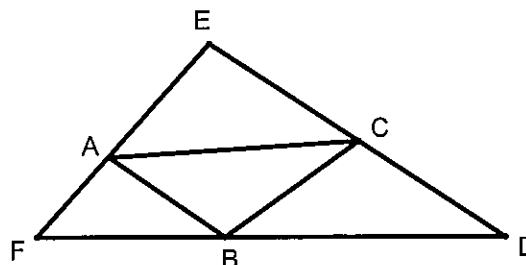
20. 2

1. Given the relation $\{(2,3),(4,k),(4,2),(5,1)\}$. The **product** of all the distinct numbers in the **range** of the relation shown is 60. Find the value of k .
2. Let $i = \sqrt{-1}$. Let k and w represent positive integers with $k < w$. If $(3i)(ki)(wi) = -12i$, find the value of k .
3. The vector $(5.1, 3.4)$ is perpendicular to the vector $(-6.55, k)$. Find the value of k . Express your answer as an **exact decimal**.
4. The sum of the infinite geometric series of real terms whose first term is k is $\frac{2k}{k+1}$. If $k > 0$ and the second term is k^2 , find the value of k . Express your answer as a common fraction reduced to lowest terms.
5. Find the length of the minor axis of the conic section whose equation is $5x^2 - 10x + 9y^2 - 72y + 104 = 0$.
6. Let $i = \sqrt{-1}$ and let k and w represent real numbers. If $(k + wi)(9 - 2i) = x + 7i$ is solved for x , then $x = 74 - 14i$. Find the value of $(k + w)$.
7. Bob, Christie, and Judy repeatedly take turns tossing a fair, standard cubical die. Bob begins, Christie always follows Bob, and Judy always follows Christie. Find the probability that Christie will be the first to toss a die that lands so that its uppermost face displays an odd number of spots. Express your answer as a common fraction reduced to lowest terms.

8. Let f be a real-numbered function. If $f(x) = \frac{\log(x-2)}{x^2-40} + \sqrt{8\sqrt{3}-x}$, then the domain of f ,

given in simplified radical form in interval notation, is $(k, a\sqrt{b}) \cup (c\sqrt{d}, g\sqrt{h}]$ where k , a , b , c , d , g , and h are positive integers. Find the value of $3k+2a+b+c+d+7g+h$.

9. In the diagram, points A , B , and C lie on $\overline{FE}$, $\overline{FD}$, and $\overline{ED}$ respectively. If $EA=10$, $AF=5$, $BF=12$, $BD=22$, $CD=16$, and $CE=19$, find the area of $\triangle ABC$. Express your answer as an improper fraction reduced to lowest terms.



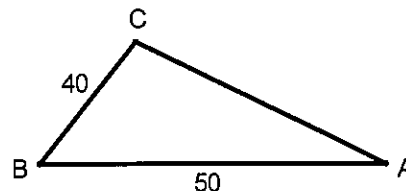
10. Find the degree of the following polynomial: $3x(x^3+5)+6(x^2(x-7)^3)^2+34x-9$.

11. There are 12 teams in a league. Each team in the league must play every other team in the league exactly 10 times in one season. Find the number of games that will be played among the teams in this league during one season.

12. Find the sum of all distinct positive values of y that satisfy the system:
$$\begin{cases} \log_9 x = \log_3 y \\ x^2 - 5y^2 = -6 \end{cases}$$

13. The first term of an arithmetic sequence is 8, and the thirteenth term of this arithmetic sequence is -28 . Find the **sum** of the other eleven terms of this arithmetic sequence.

14. In the diagram, $AB=50$ and $BC=40$. If
$$\frac{\sin(0.5(\angle CAB - \angle CBA))}{\cos(0.5(\angle ACB))} = \frac{12}{25}$$
, find the exact length AC .



15. Let $i = \sqrt{-1}$ and let k represent a real number. If $7 + ki^3 + x = 4 + 6i^2 + 3i$ is solved for x , then $x = -9 - 7i$. Find the value of k .
16. Let n represent an integer such that $0 < n < 6$. Two of the roots for x of the cubic equation $x^3 + nx^2 - (n+1)x + k = 0$ are additive inverses. Find the sum of all possible distinct values of k .
17. Katie randomly places 3 identical math, 2 identical science, and 2 identical English books on a shelf. Find the probability that the 3 math books are together on the shelf. Express your answer as a common fraction reduced to lowest terms.
18. Given that $\sin(A) = -\frac{2}{3}$ and $\pi < A < \frac{3\pi}{2}$ (where A is measured in radians). Then $\sin(2A)$ can be expressed in simplest radical form as $\frac{k\sqrt{w}}{p}$ where k and p are relatively prime integers and $p > 0$. Find the value of $(k + w + p)$.
19. Let $i = \sqrt{-1}$ and let $(x-1-2i)$, $(x-1+2i)$, and $(x-5)$ be factors of the polynomial $x^3 + kx^2 + wx + p$ where k , w , and p represent integers. Find the value of $(k + p)$.
20. In $\triangle ABC$, let $AB = c$, $AC = b$, and $BC = a$. $\angle BAC = 60^\circ$, and $(a+b+c)(a+b-c) = \frac{49}{19}ab$. If the lengths of all sides of the triangle are integers, find the smallest possible value of c .

2013 RAA

Name ANSWERS

Pre-Calculus

School _____

(Use full school name – no abbreviations)

_____ Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

1. 10

2. 1

3. 9.825 (Must be this decimal.)

4. $\frac{1}{3}$ (Must be this reduced common fraction.)

5. $2\sqrt{5}$ (Must be this exact answer.)

6. 9

7. $\frac{2}{7}$ (Must be this reduced common fraction.)

8. 91

9. $\frac{4812}{85}$ (Must be this reduced improper fraction.)

10. 10

11. 660 (Games optional.)

12. $\sqrt{2} + \sqrt{3}$ OR $\sqrt{3} + \sqrt{2}$ (Must be this exact answer.)

13. -110

14. 16

15. -10

16. -70

17. $\frac{1}{7}$ (Must be this reduced common fraction.)

18. 18

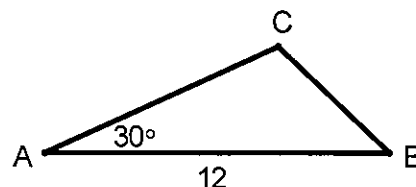
19. -32

20. 21

1. Find the value of the following limit: $\lim_{x \rightarrow 5} (2^{(9-x)})$.
2. A sequence is defined recursively by $a_{(n+1)} = 3a_n - 5$ for all integers $n \geq 1$. If $a_{10} = 16$, find the value of a_8 .
3. Let $\vec{a}$ and $\vec{b}$ represent vectors such that $\vec{a} = (3, 2)$ and $\vec{b} = (-8, 13)$. Find the **ordered pair** that represents $3\vec{a} + 5\vec{b}$.
4. Find the value of the indicated sum: $\sum_2^5 (k^3 + 3.782)$. Express your answer as an **exact decimal**.
5. Find the tenth term of the geometric sequence whose first three terms are respectively: $\frac{1024}{5}$, $\frac{512}{15}$, and $\frac{256}{45}$. Express your answer as a common fraction reduced to lowest terms.
6. Given the relation defined by $f = \{(2, 7), (5, 3), (7, 11), (19, w)\}$. Find the product of all the distinct real values of w for which f^{-1} is **not** a function.
7. Let k be a positive integer and let w be an integer such that two of the roots for x of the cubic equation $x^3 - 17x^2 + kx + w = 0$ are consecutive positive integers. Find the value of k if $|k - w|$ is a maximum.

8. The angle of elevation of the summit from the base of a ski lift is 35° . If a skier rides 1000 feet on this lift to the summit, find the number of feet in the **vertical distance** between the base of the lift and the summit. Express your answer as a **decimal** rounded to the nearest tenth of a foot.

9. In the diagram, $\angle CAB = 30^\circ$, and $AB = 12$. $AC = k\sqrt{3}$, and $BC = w$ where k and w are positive integers, and $w < 164$. Find the sum of all possible distinct values of w .



10. If $A = \begin{bmatrix} 2 & 3 \\ 1 & -3 \end{bmatrix} \bullet \begin{bmatrix} 5 & 4 \\ 1 & 7 \end{bmatrix}$, find the value of $\det(A)$.

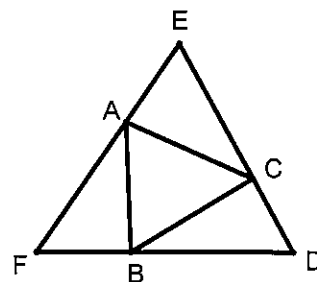
11. If $(3x - 4y + z + 2a)^2$ is expanded and completely simplified, find the sum of the numerical coefficients.

12. Let x be an integer such that $0 < x < 150$. Find the sum of all possible distinct values of x such that all three of the following conditions hold: $\tan(3x + 17)^\circ < 0$, $\sin(5x - 13)^\circ > 0$, and $\cos(2x + 3)^\circ < 0$.

13. The area of the ellipse $\frac{(x-3)^2}{36} + \frac{(y+2)^2}{k} = 1$ is 18 times the area of the circle $x^2 + y^2 = 4$.

Find the value of k .

14. In the diagram, $\triangle DEF$ is equilateral, points A , B , and C lie on $\overline{EF}$, $\overline{DF}$, and $\overline{DE}$ respectively. $\overline{EA} \cong \overline{FB} \cong \overline{DC}$. The ratio of the area of $\triangle ABC$ to the area of $\triangle DEF$ is $53:128$. If $EF = 256$, and $AE < AF$, find AF . Express your answer as a decimal rounded to the nearest tenth.



15. Given that $\sin(A) = \frac{2}{5}$ and $\frac{\pi}{2} < A < \pi$ (where A is measured in radians). Then $\sin(2A)$ can be expressed in simplest radical form as $\frac{k\sqrt{w}}{p}$ where k and p are relatively prime integers and $p > 0$. Find the value of $(k + w + p)$.
16. From a group of 8 men and 7 women, a committee of 3 is selected at random. Find the probability that the committee consists of 2 members of one sex and 1 member of the opposite sex.
17. If $\sin(x) = \frac{5}{13}$ and $\cot(y) = -\frac{3}{4}$, then there are two possible answers for $\tan(x - y)$. Find the sum of these two possible distinct answers. Express your answer as an improper fraction reduced to lowest terms.
18. Let $C(n, k) = \frac{n!}{k!(n-k)!}$ where n and k represent positive integers. Find the **sum** of all distinct values of n such that $97 < C(n, 6) < 2114$.
19. The graph of the rational function $g(x) = \frac{x^2 - 5x + 6}{x - 4}$ has a slant (oblique) asymptote which can be expressed in the form: $f(x) = mx + b$. Find the value of $(m + b)$.
20. Let x and n represent two digit positive **odd** integers for which the tens digit is non-zero and for which $n > x$. Find the largest possible value for x for which the sum of the consecutive positive **odd** integers from 1 through x is 99 less than the sum of the consecutive positive **odd** integers from x through n .

2011 RAA

Name ANSWERS

Pre-Calculus

School _____

(Use full school name – no abbreviations)

____ Correct X **2** pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

1. 1611. 42. 412. 25353. (-31,71) (Must be this ordered pair.)13. 1444. 239.128 (Must be this exact decimal.)14. 187.9 (Must be this decimal.)5. $\frac{2}{98415}$ (Must be this reduced common fraction.)15. 426. 23116. 0.8 OR .8 OR $\frac{4}{5}$ 7. 2617. $\frac{507}{112}$ (Must be this reduced improper fraction.)8. 573.6 (Must be this decimal, feet optional.)18. 469. 21619. 0 OR "zero"10. -27920. 43

1. The base of a rectangle has a length of 8. One of the whole numbers greater than 5.2 and less than 7.3 is selected at random for the height of the rectangle. Find the probability that the area of this rectangle is greater than 50. Express your answer as a common fraction reduced to lowest terms.
2. If $f(x) = |3x - 12|$, then an equation of a line of symmetry is $x = k$. Find the value of k .
3. Give the smallest possible **positive** value of x for which $\cos(2x^\circ - 60^\circ) = 0$.
4. The cubic equation whose solution set is $\{1, 5, 11\}$ can be written as $x^3 + kx^2 + wx + p = 0$. Find the value of $(k + w + p)$.
5. The first term of a geometric sequence of real terms is 405. The fifth term of this sequence is 20480. Find the third term of this sequence.
6. A circle with center at $(-6, 3)$ contains the point $(2, -12)$. An arc of the circle has measure of $35^\circ 15'$. Find the area of the sector of the circle that is bounded by two radii and the arc. Express your answer as a **decimal** rounded to the nearest tenth.
7. One of the transformations necessary to produce the graph of $y = 2x^2 + 24x - 1$ from the graph of $y = x^2$ is a vertical shift k units downward. Find the value of $|k|$.
8. The function g satisfies the functional equation $g(x) + g(y) = g(x + y) + 2xy - 8$ for every pair (x, y) of real numbers. If $g(1) = 22$, find $g(-11)$.

9. When $y^2 + my + 14$ is divided by $(y - 3)$, the quotient is $f(y)$ and the remainder is k .
When $y^2 + my + 14$ is divided by $(y - 7)$, the quotient is $g(y)$ and the remainder is w .
If $w = 2k + 1$, find the value of m .
10. All lengths of the sides of a triangle are integers. The cosine of the smallest angle of the triangle is $\frac{19}{21}$. The length of one of the adjacent sides of the triangle for that angle is 6.
Find the smallest perimeter for such a triangle.
11. From the set $\{A, B, C, D, E, F\}$, two letters are selected at random without replacement.
Find the probability that A or B or both A and B were selected. Express your answer as a common fraction reduced to lowest terms.
12. If $\ln$ is the symbol for natural logarithm, find the value of y such that
 $\ln(5 + 4y) - \ln(3 + y) = \ln(3)$.
13. In Triangle ABC , $AB = 64$, $BC = 64$, and $AC = 48$. A , B , and C are the centers of three mutually externally tangent circles whose points of tangency are D , E , and F .
Rounded to the nearest **whole number**, find the perimeter of Triangle DEF .
14. Last year UIC traced the job placement of their high school mathematics teacher candidates and found all took jobs teaching. The arithmetic mean salary of the candidates who took jobs in Chicago was \$48,600. The arithmetic mean of the rest of the candidates, who took jobs outside Chicago, was \$43,800. The overall arithmetic mean for all these candidates was \$45,000. Find the fraction of the candidates who took jobs outside Chicago. Express your answer as a common fraction reduced to lowest terms.

15. A farmer wishes to fence in a rectangular plot of ground bounded on one side by a river. He has 2200 feet of chain-link fence available. He will **not** fence the side along the river, but each of the two widths of the field perpendicular to the river will need a double layer (or “two-ply”) of fence. If the area of the rectangular plot enclosed is to be a maximum, find the number of feet in the length of the field that is parallel to the river. (Consider only the lengths of fencing used in the problem. Ignore the thickness of the double fencing in your computations.)
16. Find the absolute value in the difference of the two values of x for which the three terms $x - 2$, $x + 1$, and $4x - 8$ (in that order) will form a geometric progression?
17. (Multiple Choice) Write A if the conditions below determine a conic. Write B if the conditions below overdetermine a conic (that is, there is no conic meeting all the given conditions). Write C if the conditions below underdetermine a conic (that is, there is more than one conic that meets all the given conditions).

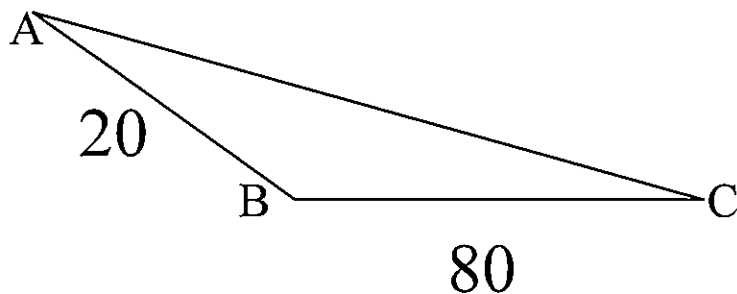
Ellipse with center at $(0, 0)$ and eccentricity of $\frac{2}{3}$

Note: Be certain to write the correct capital letter as your answer.

18. If $\sin(3x^\circ) = \frac{1}{2}$ and $133^\circ < x < 178^\circ$, find the value of x .
19. The length of the latus rectum of a hyperbola is 24.12, the eccentricity of the hyperbola is 3.62, the center of the hyperbola is at $(8.16, 19.02)$, and the principal axis of the hyperbola is parallel to the y -axis. One of the foci of the hyperbola is located at (x, y) . Find the largest possible value of y . Express your answer as a **decimal** rounded to the nearest hundredth.
20. Find the **absolute** value of the distance from the point $(1, 4, 9)$ to the plane determined by the 3 points $(5, 0, 4)$, $(1, 7, 3)$, and $(2, 8, 0)$. Express your answer as a **decimal rounded to the nearest thousandth**.

1. $\frac{1}{2}$ (Must be this reduced common fraction.)
2. 4
3. 75 (Degrees optional.)
4. -1
5. 2880
6. 88.9 (Must be this decimal.)
7. 73
8. -470
9. -16
10. 16
11. $\frac{3}{5}$ (Must be this reduced common fraction.)
12. 4
13. 84 (Must be this whole number.)
14. $\frac{3}{4}$ (Must be this reduced common fraction.)
15. 1100 (Feet optional.)
16. 4
17. C (Must be this capital letter.)
18. 170 (Degrees optional.)
19. 22.63 (Must be this decimal.)
20. 1.028 (Must be this decimal.)

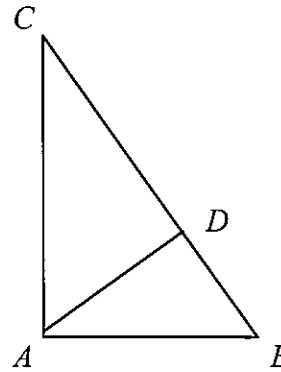
1. $\begin{bmatrix} 3 & 2 \\ 4 & 6 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \end{bmatrix} = \begin{bmatrix} k \\ w \end{bmatrix}$. Find the value of $(2k + 3w)$.
2. Given the set: $\{\log_2(16), \log_3(16), \log_4(16), \log_5(16)\}$. If one of the 4 members of the set is drawn at random, find the probability that the member drawn could represent a positive integer. Express your answer as a common fraction reduced to lowest terms.
3. Find the ordered pair that represents the sum of the following two vectors: $(-5, 6)$ and $(17, 7)$.
4. Let n be a positive integer. For $n \geq 1$, $x_{(n+2)} = x_{(n)} + x_{(n+1)}$. If $x_1 = 3$ and $x_2 = 1$, find the value of x_{10} .
5. Matrix $[A]$ contains 23 rows and 13 columns. Matrix $[B]$ contains 13 rows and 92 columns. Find the number of rows in the product matrix $[A][B]$.
6. In $\triangle ABC$ as with side-lengths as shown, $\angle ABC = 120^\circ$. Find the exact length of $\overline{AC}$.



7. A woman is sitting in the stands behind one end zone of a football field. She observes, in the same vertical plane with her position, two players standing on their own goal lines, exactly 100 yards apart. Looking at the spots on the ground where they are standing, a player at one goal line is at an angle of depression of 15° and a player at the other goal line is at an angle of depression of 8° . Assuming the football field is horizontal, find the **number of feet** in the vertical height of the woman's eye above the horizontal plane of the football field. Express your answer as a **decimal** rounded to the nearest hundredth of a foot.

8. If three real geometric means are inserted between 15 and $\frac{5}{27}$, find the value of the middle term. Express your answer as an improper fraction reduced to lowest terms.

9. In the diagram $\triangle CAB$ is a right triangle with $\angle CAB = 90^\circ$. Point D is on $\overline{BC}$ such that $\overline{AD} \perp \overline{BC}$. Let the lengths of $\triangle CAB$ be a , b , and c with these lengths respectively opposite the angles with vertices at A , B , and C . If $b \sin C (b \cos C + c \cos B) = 41$, and if the length of $\overline{AD}$ is an integer, find the largest possible value of AD .



10. The roots of $x^3 + kx^2 - 59x + 420 = 0$ are 5, -7 , and 12. Find the value of k .
11. When $(2x - 3)^8$ is expanded and completely simplified, find the numerical coefficient of x^5 .
12. The first term of an infinite geometric sequence of real terms is 225, and the fourth term of this geometric sequence is 14.4. Find the sum of the terms of this infinite geometric sequence.
13. The sum of the terms of an arithmetic sequence with 13 terms and common difference of 3 is found to be 286. Find the thirteenth term of this sequence.
14. Let there be k objects, each of weight w . When these objects are weighed in pairs, the sum of the weights of all possible pairs is 770. When these objects are weighed in groups of four, the sum of the weights of all possible groups of four is 9240. Find the value of $(k + w)$.

15. Let $i = \sqrt{-1}$. Find the exact value of $|\sqrt{6} + i\sqrt{8}|$. Write your answer in simplified radical form.
16. Tom is a gambler. He has a pair of fair, standard cubical dice. He selects two different numbers in advance from the face numbers 1, 2, 3, 4, 5, and 6. He then rolls the pair of dice. If both numbers showing match his two selections, he wins \$108. If exactly one number showing matches one of his two selections, he neither wins nor loses. If neither number showing matches either of his two selections, he loses \$54. Find Tom's mathematical expectation each time he rolls the pair of dice. Express your answer by including the word "lose" or the word "win." For example, your answer might be "lose \$2" or might be "win \$7."
17. Let D be the point that is $\frac{1}{3}$ of the way from B to A for equilateral triangle $\triangle ABC$. Let P be a point on $\overline{BC}$ such that $AP + PD$ is the smallest possible value. If the perimeter of $\triangle ABC$ is 63, find the exact value of $(AP + PD)$.
18. Let $i = \sqrt{-1}$ and let k and w represent real numbers. If $(k + wi)(4 + i) = 33 - 13i$ then find the value of $(k + w)$.
19. Find the acute angle between the normal vectors of the following two planes:
- $$3x + 2y + 5z + 4 = 0 \text{ and } -2x - 3y + 6z - 2 = 0$$
- Round your answer to the nearest minute and then express your answer in degrees and minutes in the form: $k^\circ w'$.
20. Points A , B , C , and D are the vertices of a rectangle. Point E is in the interior of this rectangle. $AE = 8$, $EC = 21$. The lengths of $\overline{DE}$, $\overline{BE}$, and $\overline{BC}$ are integers. If $DE < BE$, find the largest possible value for the length of $\overline{BC}$.

2009 RAA

Name ANSWERS

Pre-Calculus

School _____

(Use full school name – no abbreviations)

_____ Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

1. 68

2. $\frac{1}{2}$ (Must be this reduced common fraction.)

3. (12,13) (Must be this ordered pair.)

4. 97

5. 23

6. $20\sqrt{21}$ (Must be this exact simplified radical.)

7. 88.67 (Must be this decimal, feet optional.)

8. $\frac{5}{3}$ (Must be this reduced improper fraction.)

9. 4

10. -10

11. -48384

12. 375

13. 40

14. 18

15. $\sqrt{14}$ (Must be this exact simplified radical.)

16. Lose 18 (Must have "lose", not -18, \$ or dollars optional.)

17. $7\sqrt{13}$ (Must be this exact simplified radical.)

18. 2

19. $65^{\circ}21'$ (Must be in the form Degrees and minutes.)

20. 19